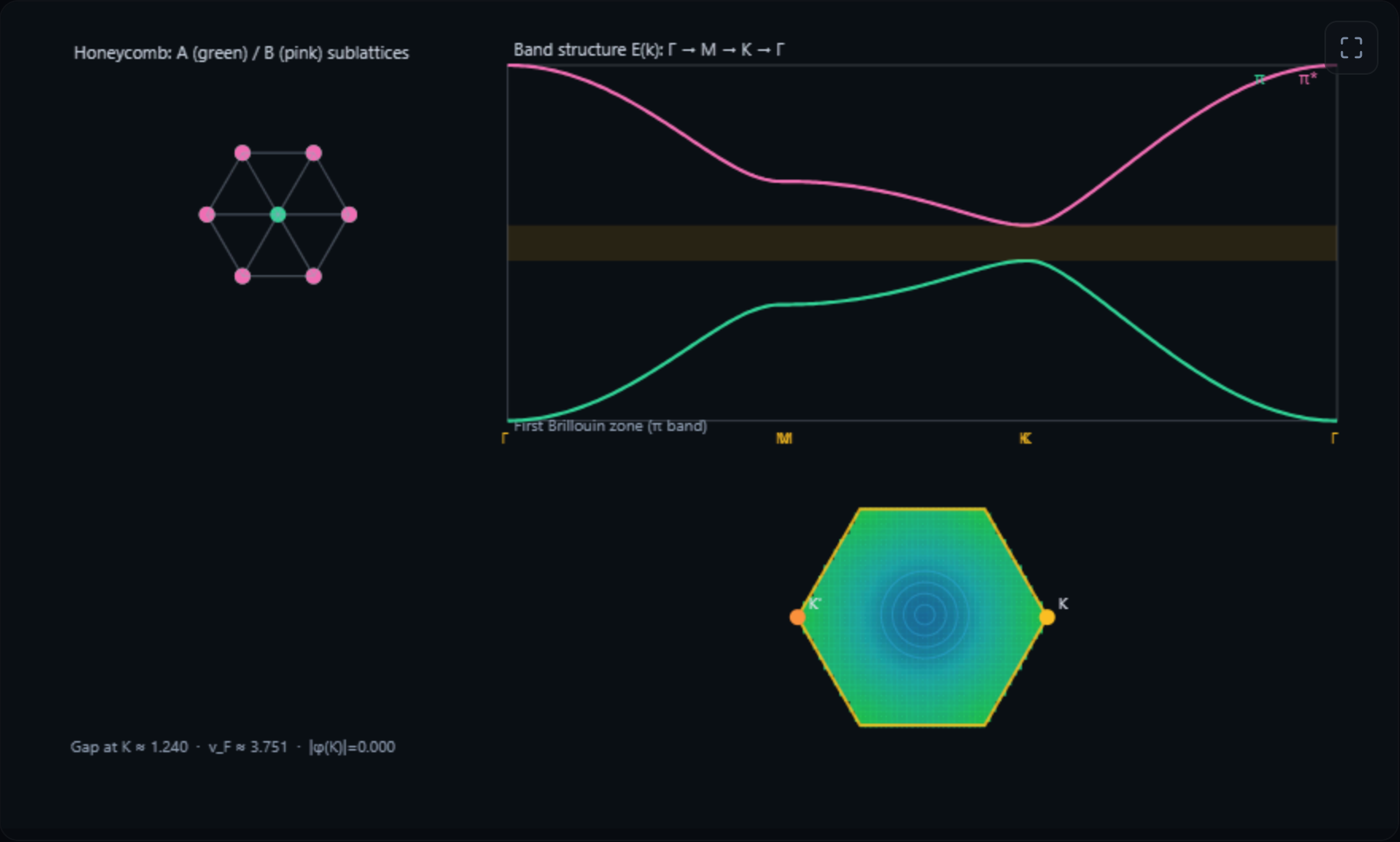


Graphene Tight-Binding Band Structure

Honeycomb π tight binding: $E(k)$ along Γ – M – K – Γ , Dirac cones at K and K' when $\Delta=0$, gap 2Δ from sublattice stagger, and a Brillouin-zone energy map.



Δ

0.62

About this model

Graphene’s low-energy physics is captured by nearest-neighbor tight binding on a honeycomb lattice with two sublattices A and B. With nearest-neighbor vectors $(0,a)$ and $(\pm\sqrt{3}a/2,-a/2)$, the structure factor $\varphi(k) = e^{iak_y} + 2e^{\{-iak_y/2\} \cos(\sqrt{3} a k_x/2)}$ vanishes at the Dirac points K and K’, producing linear π and π^* bands $E_{\pm}(k) = \pm\sqrt{(\Delta^2 + \tau^2|\varphi(k)|^2)}$ when sublattice onsite energies $\pm\Delta$ are included. Pristine graphene ($\Delta = 0$) has touching cones with Fermi velocity $v_F \approx (3/2)\tau a$; a finite stagger opens a gap 2Δ at K and K’ because $|\varphi| = 0$ there. This simulator plots $E(k)$ along the standard path $\Gamma \rightarrow M \rightarrow K \rightarrow \Gamma$, colors the π band in the first Brillouin zone, and marks K and K’. It is a two-band, single- π -orbital model — no next-nearest hopping, trigonal warping, or spin–orbit coupling.

Who it's for: Advanced undergraduate or graduate students in condensed-matter or solid-state physics after Bloch’s theorem and before the quantum Hall effect or bilayer graphene.

KEY TERMS

- Graphene
- Honeycomb lattice
- Tight-binding model
- Dirac cone
- Brillouin zone
- Band gap
- Fermi velocity

How it works

Graphene tight-binding band structure on a honeycomb lattice: π and π^* bands along Γ –M–K– Γ , Dirac cones at K and K’ when $\Delta=0$, and a tunable gap from sublattice staggered potential Δ .

Frequently asked questions

Why are there two Dirac points K and K’?

The honeycomb Brillouin zone has two inequivalent corners related by inversion. Time-reversal symmetry pairs them; each hosts a linear band crossing when $\Delta = 0$.

What opens the band gap?

A staggered sublattice potential (e.g. on h-BN substrate or in biased bilayer) sets $\epsilon_A = -\Delta$ and $\epsilon_B = +\Delta$. At K and K’ where $|\varphi(k)| = 0$, the gap is 2Δ independent of hopping τ .

What is $\phi(k)$?

It is the sum of phase factors from the three nearest-neighbor hoppings between A and B sites. Its magnitude sets the band splitting away from the Dirac points.

Is this quantitative for real graphene?

Only qualitatively. Real graphene needs fitted τ and $a \approx 1.42 \text{ \AA}$, includes trigonal warping and interactions; this page teaches the canonical two-band Dirac physics.

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Step ΔH° table and enthalpy ladder; lattice energy $U = \Sigma(\text{gas-ion steps}) - \Delta H_f^\circ$.